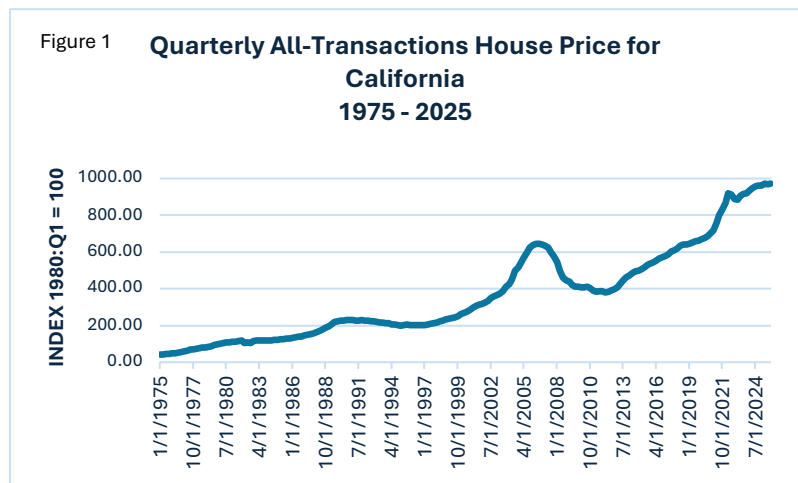
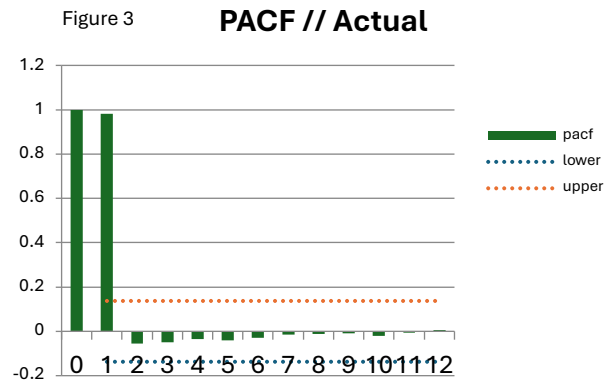
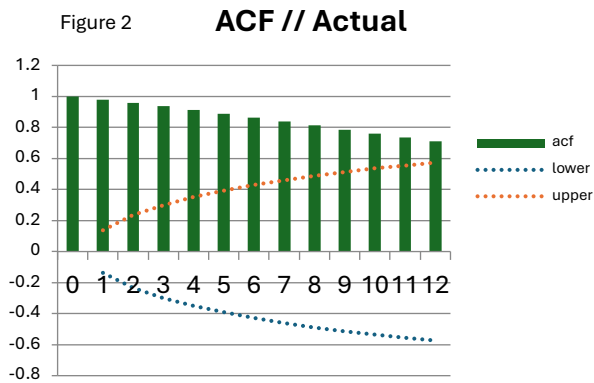


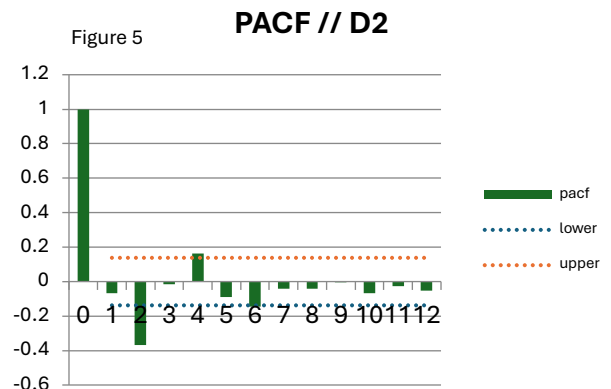
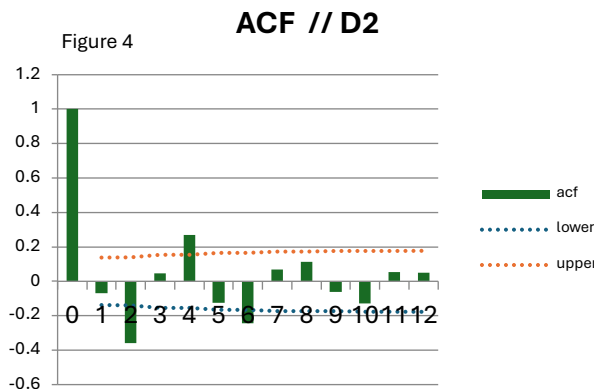
We have implemented three new forecasting equations to aid in the outlook for the cost of housing in California in the next year. In doing so, we needed to settle for the difference in the observed data to apply the ARIMA and SARIMA models. To start out, we needed to form a stationary series as this is a critical assumption when calculating the output. We wouldn't be able to do it with the raw data, as the time-series shows an upward trend over time. There was also the presence of cycles in the data, and the observed mean values are not constant (Figure 1).



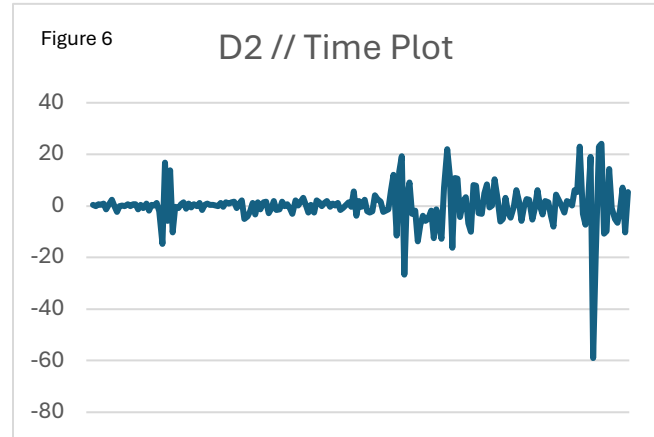
I ran a correlogram test to visually inspect the autocorrelations of the actual data. This is to see the correlation between an observed and lagged value. In the correlogram of the actual data (Figure 2), the ACF shows a slow decline for the twelve lags and all spiking out of the interval. This means that there is high autocorrelation between observation and lag, which is dependent on time itself. This would imply that there is a trend occurring with the time-series if we were not seeing a line graph of data. The second correlogram to compliment this is the PACF (Figure 3), which shows a spike at lag one but sharp drops with all autocorrelation values staying within the interval.



To pursue my forecast, I implemented a lag of one in the time-series to meet the conditions of stationarity. In the first difference (D1), there have been improvements in the ACF as the autocorrelation of the values has lowered. Still, there are spikes shown above the desired intervals. The PACF now has spikes at lag three and five. This provides information that there is a correlation between the observed value and distant lag when accounting for the lags between them. When we move to the second difference (D2), it improves on the first one with majority of the lags lowering sharply in the ACF and PACF (Figure 4 & 5). However, there are still spikes that are present beyond the intervals, which indicates that data is still not fully stationary. The line plot of the second difference concludes this also as there are large swings throughout the data. (Figure 6)



Out of both, I decided on using the second difference for the ARIMA model as the autocorrelation levels are lower and the swings are more consistent for the most part (Figure 6). When deciding which aspects to implement in the model, I went with an ARIMA (1,2,1) as the AFC and PACF show



similar output. The two correlograms show no gradual declines in the autocorrelations and show similar spikes at the lags. So, I implemented both the non-seasonal Autoregressive (AR) and Moving average (MA), with a non-seasonal difference of two.

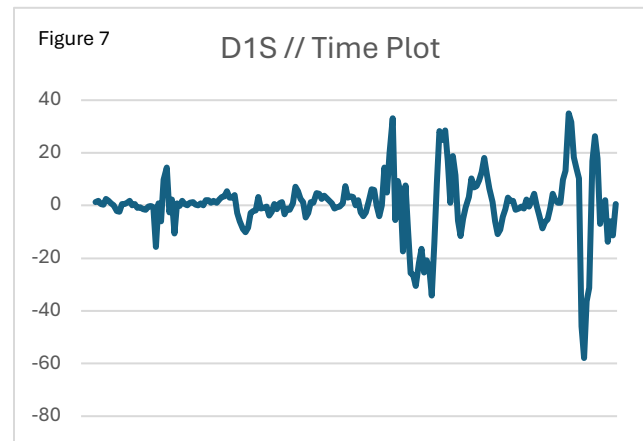
When measuring how close the model is to the actual data, the Mean Absolute Deviation (MAD) is at 4.36 and the Root Mean Squared Error (RMSE) is 7.77. Another measure for how the model fits is the Akaike Information Criterion (AIC). This accounts for the added parameters in the ARIMA model and the results are 832.13. These measures will be used to compare with the SARIMA models later to determine if lower values can be possible. In terms of seeing if the residuals still correlate between each other, I used the Ljung-Box test. The output shows that the p-value is less than the alpha of 0.05, which rejects the null hypothesis. The results conclude that the residuals are not completely static and there is still useful data that is not being accounted for.

The next set of models applies seasonality to ARIMA, which could account for any non-realized data that was missed earlier. The first seasonal method applies the general purpose “airline” model, which is SARIMA (0,1,1) x (0,1,1). This implements one non-seasonal and seasonal difference and MA term. When measuring the fit, the MAD is 4.88 and the RMSE is 8.37. Since these values are larger than the ARMA model, this performs worse in comparison. The AIC

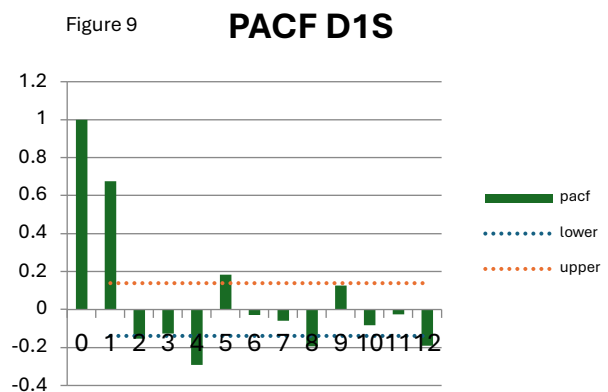
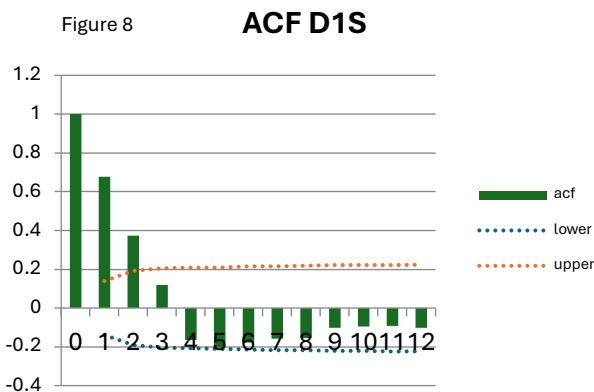
results show a value of 851.57, which is higher than the previous AIC. This can be due to having more parameters, as additional ones penalize the score. The output from the Ljung-Box Test shows that residuals are not mostly white noise. The p-value is higher than ARMA's but still falls below the alpha of 0.05. Which results in rejecting the null as there is still data not accounted for.

The second seasonal model applied is a SARIMA (1,1,0) x (1,1,0), uses one seasonal and non-seasonal AR term and difference. To elaborate on how the actual observations are shaped into the model. The data is calculated by the seasonal difference from the first difference, to form a more stationary series (Figure 7).

The AFC output from this shows a decline in the autocorrelations of each lag, except for three and four. There are no present spikes



after lag two as the rest stay within the intervals. (Figure 8) The PACF shows a majority staying within the interval with noticeable spikes in lags four, eight, and twelve. (Figure 9) I decided on an AR term for both seasonal and non-seasonal as these correlograms tilt more into the description.



The output from the SARIMA shows an improved fit as the MAD decreases to 4.85, the RMSE lowers to 8.17, and the AIC is 821.09. However, when testing with the Ljung-Box Test, the results show that the residuals are not white noise. The p-value is the highest out of all the models but is still lower than the alpha of 0.05. So, the null must be rejected, and this concludes there is still some correlation between the lags.

Figure 10

Model Name	RMSE	MAD	Theil's U	AIC
ARIMA	7.769	4.358	3.628	832.131
SARIMA 1 (Airline)	8.367	4.876	3.907	851.470
SARIMA 2	8.172	4.849	3.817	821.090

When comparing all the models applied to the all-transaction house price data for California, the summary found that the second SARIMA

model performs the best (Figure 10). The main reason for this decision is the improved AIC as it is the lowest out of all of them. RMSE and MAD fall in between the ARIMA and the first SARIMA models. When comparing the Theil's U measure, it also falls in between the two also. The p-values of the second SARIMA is also the highest, which tells me that this model came the closest to achieving white noise. When compared to the Naïve one, all the methods used are

above the Theil's U as they reach near four. This shows that the Naïve one still a reasonable candidate to add into the forecasting process. If we were to apply the second SARIMA in a forecast for the year ahead, it projects a steady rise in the house price index for 2026 (Figure 11).

Figure 11

